

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Thursday, 17 September 2026, 8:42 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:44 AM**Time taken** 1 min 44 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3  int findDuplicate(const int* arr, int n) {
4      if (n <= 1) return -1;
5      int tortoise = arr[0];
6      int hare = arr[0];
7      do {
8          tortoise = arr[tortoise];
9          hare = arr[arr[hare]];
10     } while (tortoise != hare);
11     tortoise = arr[0];
12     while (tortoise != hare) {
13         tortoise = arr[tortoise];
14         hare = arr[hare];
15     }
16     return tortoise;
17 }
18 int main() {
19     int n;
20     if (scanf("%d", &n) != 1) return 0;
21     int* arr = (int*)malloc(n * sizeof(int));
22     for (int i = 0; i < n; i++) {
23         if (scanf("%d", &arr[i]) != 1) {
24             free(arr);
25             return 0;
26         }
27     }
28     int duplicate = findDuplicate(arr, n);
29     printf("%d\n", duplicate);
30     free(arr);
31     return 0;
32 }
33

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)